Table 1: Generation with observed variables and known structure: mean and sample standard deviation across three seeds. Each entry shows the mean above its ± standard deviation. Validity and coverage are percentages. Bold marks the best mean within each DGP and metric.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑
CD	13.55 ± 2.94	0.1467 ±0.0713	40.00 ± 5.06	83.22 ± 4.51	0.0025 ±0.0019	78.52 ± 6.64	78.12 ± 8.62	0.0062 ±0.0036	74.35 ± 3.97	85.55 ± 3.85	0.0034 ±0.0018	70.49 ± 5.44	80.65 ± 7.18	0.0072 ±0.0038	73.08 ± 3.27
Compositional CD	75.50 ± 1.96	0.1065 ±0.0072	84.78 ± 3.61	38.54 ± 1.76	0.0139 ±0.0014	31.25 ± 4.37	31.74 ± 1.32	0.0368 ±0.0061	27.67 ± 4.42	21.21 ± 3.04	0.0497 ±0.0037	16.06 ± 5.35	20.93 ± 2.47	0.1054 ±0.0010	20.04 ± 3.64
Additive mean†	0.00 ± 0.00	0.1174 ±0.0093	0.00 ± 0.00	**100.00** ± 0.00	0.0088 ±0.0026	0.55 ± 0.96	0.00 ± 0.00	0.1711 ±0.0024	0.00 ± 0.00	0.00 ± 0.00	0.0716 ±0.0104	0.00 ± 0.00	0.00 ± 0.00	0.3416 ±0.0106	0.00 ± 0.00
Additive + residual†	0.18 ± 0.06	0.2322 ±0.0041	0.65 ± 0.39	7.81 ± 1.53	0.1407 ±0.0076	6.17 ± 6.23	0.00 ± 0.00	0.2037 ±0.0008	0.00 ± 0.00	1.35 ± 0.50	0.1523 ±0.0089	0.86 ± 0.69	0.18 ± 0.20	0.3266 ±0.0053	0.10 ± 0.10
Fixed composition + residual†	0.13 ± 0.03	0.2353 ±0.0032	0.50 ± 0.20	0.03 ± 0.06	0.2327 ±0.0060	0.00 ± 0.00	0.02 ± 0.03	0.2014 ±0.0141	0.00 ± 0.00	0.07 ± 0.07	0.2548 ±0.0161	0.00 ± 0.00	0.00 ± 0.00	0.3798 ±0.0096	0.00 ± 0.00
Rep4Ex-inspired + residual†	0.85 ± 0.74	0.2714 ±0.0658	2.87 ± 2.45	3.68 ± 3.83	0.2490 ±0.1286	3.06 ± 3.89	0.23 ± 0.35	0.2732 ±0.1037	0.02 ± 0.03	0.39 ± 0.32	0.2621 ±0.1050	0.02 ± 0.03	0.03 ± 0.06	0.3648 ±0.1766	0.03 ± 0.06
CG: sparse factorized ($s = 1$)	46.93 ± 7.34	0.1027 ±0.0106	74.85 ± 3.03	5.40 ± 2.81	0.0776 ±0.0026	1.84 ± 1.49	8.92 ± 2.17	0.1280 ±0.0034	4.07 ± 2.65	3.43 ± 1.08	0.1096 ±0.0114	0.75 ± 0.20	4.07 ± 0.63	0.2045 ±0.0129	3.52 ± 0.81
CG: dense factorized ($s = 1$)	55.43 ± 6.43	0.1073 ±0.0057	75.58 ± 2.85	7.67 ± 4.64	0.0692 ±0.0026	2.86 ± 2.14	3.45 ± 0.70	0.1266 ±0.0039	0.65 ± 0.46	2.86 ± 1.90	0.1073 ±0.0058	0.80 ± 0.33	1.69 ± 1.21	0.2084 ±0.0197	1.79 ± 1.71
CG: dense joint ($s = 1$)	1.47 ± 1.22	0.5295 ±0.0298	4.27 ± 3.54	15.35 ± 11.25	0.0359 ±0.0074	6.74 ± 6.44	13.22 ± 8.30	0.0745 ±0.0064	5.49 ± 4.16	14.75 ± 11.00	0.0526 ±0.0078	4.57 ± 4.76	6.18 ± 6.07	0.1360 ±0.0138	4.82 ± 4.66
CG: sparse factorized ($s = 2$)	76.13 ± 5.91	0.0530 ±0.0196	61.22 ± 1.04	24.82 ± 11.92	0.0161 ±0.0007	7.42 ± 7.76	35.74 ± 8.97	0.0183 ±0.0024	13.85 ± 10.90	19.48 ± 10.72	0.0383 ±0.0099	4.87 ± 2.69	36.04 ± 4.75	0.0758 ±0.0052	30.99 ± 2.84
CG: dense factorized ($s = 2$)	**82.85** ± 5.68	0.0633 ±0.0116	56.77 ± 3.14	29.61 ± 16.03	0.0116 ±0.0045	9.83 ± 7.13	26.16 ± 11.93	0.0194 ±0.0003	2.67 ± 1.30	17.01 ± 14.70	0.0423 ±0.0048	1.14 ± 1.15	5.70 ± 4.56	0.0752 ±0.0110	4.64 ± 4.15
CG: dense joint ($s = 2$)	0.45 ± 0.78	0.5508 ±0.0336	1.42 ± 2.41	18.20 ± 26.55	0.0111 ±0.0080	9.18 ± 15.52	34.49 ± 25.69	0.0143 ±0.0037	13.77 ± 12.87	25.72 ± 20.61	0.0241 ±0.0018	6.35 ± 6.83	14.19 ± 19.63	0.0394 ±0.0117	9.52 ± 12.63
SEDGE: no weighting	12.00 ± 0.15	0.2198 ±0.0133	29.95 ± 3.03	8.98 ± 1.36	0.4006 ±0.0066	20.82 ± 5.01	8.97 ± 1.34	0.3650 ±0.0061	17.61 ± 3.34	8.92 ± 1.34	0.4248 ±0.0069	17.12 ± 2.88	2.46 ± 0.49	0.6294 ±0.0069	4.10 ± 0.91
SEDGE-LGD ($s = 0.5$)	38.07 ± 2.28	0.1525 ±0.0028	70.00 ± 2.51	44.11 ± 5.68	0.1343 ±0.0121	64.81 ± 6.63	31.92 ± 2.40	0.2048 ±0.0153	49.61 ± 5.06	47.90 ± 7.27	0.1267 ±0.0251	60.01 ± 10.11	23.67 ± 2.13	0.3420 ±0.0225	35.40 ± 1.17
SEDGE-LGD ($s = 1$)	56.80 ± 4.00	0.0996 ±0.0107	81.08 ± 5.86	78.24 ± 2.88	0.0104 ±0.0019	79.17 ± 1.27	60.17 ± 6.17	0.0711 ±0.0199	64.49 ± 5.27	79.57 ± 4.48	0.0147 ±0.0063	70.05 ± 4.07	66.37 ± 2.01	0.1153 ±0.0295	66.60 ± 1.44
SEDGE-LGD ($s = 2$)	39.98 ± 19.14	0.0272 ±0.0095	57.12 ± 18.38	77.07 ± 7.78	0.0302 ±0.0064	62.42 ± 7.84	71.89 ± 4.58	0.0302 ±0.0024	50.91 ± 3.46	79.46 ± 4.99	0.0328 ±0.0075	55.27 ± 9.85	76.25 ± 3.09	0.0516 ±0.0229	56.77 ± 3.07
SEDGE-LWS	77.55 ± 3.96	**0.0225** ±0.0183	**89.97** ± 1.11	91.98 ± 1.33	**0.0018** ±0.0013	**87.37** ± 3.93	**91.26** ± 1.27	**0.0024** ±0.0021	**84.29** ± 2.46	**91.55** ± 1.08	**0.0030** ±0.0007	**82.23** ± 3.58	**87.39** ± 3.02	**0.0016** ±0.0004	**78.86** ± 1.11

Standard deviations use $n - 1$ in the denominator ($n = 3$), not standard errors. Displayed zeros may reflect rounding. Validity requires both specifications and root support; J-MMD measures joint (X, Z) fidelity; coverage uses the reference-based fixed-radius criterion. †Paper-inspired adaptations, not faithful reproductions. Additive mean is a deterministic control, not a posterior sampler. All three original LGD scales and CG scales 1 and 2 are shown; the expanded LGD sweep is excluded. Weighted Discrete is a relaxed Assumption-3 control (conditional exclusion 2.5%/5%); continuous settings meet the working 1% band-conditional criterion. This results-informed selection does not establish uniform superiority. Boldface does not imply statistical significance.